Javad Komijani

Curriculum Vitae

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Professional Summary

Theoretical and computational physicist working at the intersection of lattice field theory, mathematical physics, and machine learning. I develop research ideas end to end, from mathematical formulation and algorithm design to implementation and large-scale computation. My distinctive expertise, rooted in effective field theories (EFTs), is identifying the collective variables that drive large-scale behavior in complex systems, and using them to interpret data and to construct efficient quantitative models with controlled uncertainties. Notably, I independently developed the theory, EFT model, and statistical analysis for precision quark masses, with results incorporated into the world averages of the Particle Data Group. Alongside this, I work on asymptotic analysis and perturbation theory, including divergent series and renormalons in QCD, and nonlinear eigenvalue problems.

Since 2021, my research has focused on machine learning for physics. I combine the EFT perspective with symmetries and equivariance to develop efficient generative models for lattice gauge theories, including normalizing flows and diffusion models on Lie groups, and I am interested in how these principles can inform interpretability, distillation, and the understanding of scaling behavior in machine learning.

Research impact: 27 peer-reviewed publications, h-index 20, 3500+ citations (Google Scholar, Nov 2026).

Flagship Research Contributions

- **Precision lattice QCD and Standard Model parameters:** Contributed to high-precision determinations of quark masses, meson decay constants, and hadronic matrix elements through the MILC, Fermilab Lattice, HPQCD, and TUMQCD collaborations. These results are incorporated into global reference averages, including FLAG and the PDG [PRD 98, 054517 (2018); PRD 98, 074512 (2018)].
- **Effective field theory for multi-scale QFT:** Developed and applied effective field theory frameworks (Symanzik EFT, ChPT, HQET) to systematically connect lattice simulations with continuum QCD observables, enabling controlled multi-scale analyses.
- **MRS scheme:** Developed the minimal renormalon-subtracted (MRS) scheme, providing a systematic treatment of renormalon ambiguities in perturbative QCD without introducing an arbitrary subtraction scale, enabling stable multi-scale analyses and high-precision heavy-quark mass determinations [PRD 97, 034503 (2018)].
- **Diffusion models on Lie groups:** Developed an implicit score-matching framework for Lie groups and predictor–corrector samplers with Hamiltonian-dynamics and Langevin correctors for SU(N) lattice gauge theories [JHEP (accepted), arXiv:2605.06134]; introduced a noise schedule on Lie groups that yields a linear decay of the Wilson action with diffusion time [arXiv:2605.17326]; co-developed group-equivariant diffusion models for lattice field theory [JHEP 07, 213 (2026)].
- **Normalizing flows for lattice field theory:** Identified the multi-scale structure near critical points as a source of poor scaling of generative models and proposed architectures inspired by effective field theory [PoS Lattice 2022, 019]; introduced SVD-based gauge-equivariant flows for SU(N) together with a parametrization of SU(3) eigenvalues [PoS Lattice 2024, 050]; reformulated two-dimensional SU(N) gauge theory as independent plaquettes learned by small spline flows, combined with an exact

reweighting [Phys. Rev. D (accepted), arXiv:2610.09147].

- **Scientific machine learning infrastructure:** Developed GPU-accelerated software frameworks for diffusion models, normalizing flows, and differentiable scientific computing (lattice_ml & normflow).

Mathematical & Asymptotic Methods

- **Nonlinear eigenvalue problems:** Introduced analytical and numerical methods for nonlinear eigenvalue problems arising in nonlinear systems with separatrix structure [J. Phys. A 47, 235204 (2014); J. Phys. A 54, 465202 (2021)].
- **Hyperasymptotics:** Developed and applied hyperasymptotic and resummation techniques for perturbative expansions in QFT, enabling controlled treatment of divergent series and renormalon ambiguities [J. Phys. A 47, 235204 (2014); JHEP 08, 062 (2017); PRD 97, 034503 (2018)].

Education

2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*
Advisors: Claude Bernard and Carl M. Bender
- Developed EFT-based frameworks and Bayesian inference methods for lattice QCD.
- Performed precision determinations of hadronic observables in lattice QCD.
- Developed a novel definition of eigenvalue problems for nonlinear systems arising in physics.

2009 **MSc in Electrical Engineering**, *University of Tehran*
- Focus: electromagnetic waves and fields, signal processing, statistical analysis.

2006 **BSc in Electrical Engineering**, *University of Tehran*

Professional Experience

2021–2026 **Postdoctoral Researcher**, *ETH Zurich*
- Developed diffusion and score-based generative models on Lie groups for lattice gauge theory.
- Built differentiable computing frameworks using PyTorch (ODE solvers and linear algebra).
- Integrated normalizing flow modules into the EU interTwin Digital Twin Engine.
- Developed distributed training infrastructure for large-scale lattice simulations.
- Mentored PhD students and co-supervised MSc, BSc, and internship projects in scientific machine learning.

2019–2020 **Postdoctoral Researcher**, *University of Tehran*
- Performed precision determinations of hadronic form factors.

2017–2018 **Postdoctoral Researcher**, *University of Glasgow*
- Computed hadronic form factors and quark condensates.

2015–2017 **Postdoctoral Researcher**, *Technical University of Munich*
- Performed precise determinations of quark masses and decay constants.

Open-Source Scientific ML Software

lattice_ml **PyTorch library for ML-assisted sampling in lattice field theory (v0.2.0)**
- Diffusion models on Lie groups with predictor–corrector sampling; flow matching for SU(N).
- HMC; adjoint, Lie-group, and symplectic ODE solvers; AD-safe SVD and eigendecomposition.
- Gauge-equivariant layers, Lie-group coordinates, random-matrix sampling; distributed training.

normflow PyTorch package for normalizing flows on Lie groups and lattice gauge theory, including symmetry-preserving transformations and flow-based sampling methods (v3.0.0).

Notebooks Reproducible notebooks for arXiv:2610.09147.

interTwin Contributed normalizing-flow-based ML modules to the EU interTwin Digital Twin Engine.

Technical Skills

Machine Learning: PyTorch, Diffusion Models, Normalizing Flows, Score Matching, Generative Modeling, Distributed Training.

Programming: Python, C++, Cython, Bash.

Scientific Computing: NumPy, SciPy, Pandas, SQL, Monte Carlo Methods, Bayesian Inference, Statistical Modeling, Differentiable ODE Solvers.

Infrastructure: Linux, Git, GPU computing, HPC and Supercomputing Environments.

Mathematics: Differential Equations, Linear Algebra, Asymptotic Analysis, Optimization.

Mentoring and Co-supervision

PhD
- Octavio Vega (2024–2026) — Group-equivariant diffusion models for lattice field theory.
- Ben Izett (2026) — Fourier acceleration for SU(N) lattice gauge theory.

MSc
- Lara Turgut (2025–2026) — Gauge-equivariant U-Net for diffusion models in lattice gauge theory.
- Ahmed Chahlaoui (2024–2025) — Autoregressive models for lattice configurations.
- Antoine Misery (2024) — Continuous-time generative models for lattice fields.
- Flavia Giehr (2024) — Continuous normalizing flows with adjoint methods.
- Klemen Kersic (2023) — Rational-quadratic splines for normalizing flows.
- Lea Segner (2023) — Simulated tempering for topological freezing.

BSc
- Nicolas Müller (2025) — Path integral formulation of QM and Monte Carlo Simulations.

Intern
- Elias Nyholm (2023) — CNN architectures in arbitrary dimensions.

Teaching Experience

ETH Zurich
- **Substitute lecturer & head teaching assistant:** QM I & II (2024).
- **Proseminar mentoring and supervision:** General Relativity (Spring 2025), Riemann Surfaces in Mathematical Physics (Spring 2023), Systems out of Equilibrium (Fall 2022), Quantum Information (Spring 2022), Theoretical Physics (Fall 2021).

U Tehran
- **Organizer and lecturer:** Workshop on lattice field theory for students (Fall 2019).

U Glasgow
- **Teaching assistant:** C Programming under Linux (Spring 2018).

TU Munich
- **Teaching assistant:** Particle Physics (Spring 2016) and QFT (Fall 2015).

Wash. U.
- **Teaching assistant:** Statistical Mechanics (Spring 2015), Quantum Mechanics (Fall 2013), Physics I (Fall 2012), Physics II (Spring 2011), Special Relativity (Spring 2011).

Invited Presentations

ECT* 2023 Machine learning for lattice field theory and beyond, ECT* in Trento, Italy, 26–30 Jun, 2023.

ETH 2022 Efficient simulations on GPU hardware, ETH Zurich, 24–27 Oct, 2022.

$g-2$ 2022 Fifth Plenary Workshop of the Muon $g-2$ Theory Initiative, Edinburgh, 5–9 Sep, 2022.

CKM 2018 10th International Workshop on the CKM Unitarity Triangle, Heidelberg, 17–21 Sep, 2018.

Latt. 2018 36th International Symposium on Lattice Field Theory, East Lansing, 22–28 Jul, 2018.

Professional Service

Refereeing Referee for Physical Review D and other physics journals (10+ reviews since 2016), including machine-learning papers.

Organization **Co-organizer of international and local workshops:**

- ZPW 2026 — Machine Learning Techniques for Particle Physics.
- QCD 2025 — Symposium on QCD Factorization.
- ML Lattice 2025 — Machine learning approaches in Lattice QCD.
- UT 2019 — Lattice QCD workshop.
- TUM 2017 — EFT and Lattice Field Theory Summer School.

Selected Publications

Generative Models

- J. Komijani, "Sampling SU(N) gauge theory on a 2D lattice from independent plaquettes via holonomies and corner reweighting," Phys. Rev. D (accepted), [arXiv:2610.09147].
- J. Komijani, "Noise scheduling and linear dynamics in diffusion models on Lie groups," [arXiv:2605.17326].
- J. Komijani, M. K. Marinkovic and L. Turgut, "Diffusion model for SU(N) gauge theories," JHEP (accepted), [arXiv:2605.06134].
- O. Vega, J. Komijani, A. El-Khadra and M. K. Marinkovic, "Group-equivariant diffusion models for lattice field theory," *JHEP 07, 213 (2026)*.
- A. Manzi *et al.*, "interTwin: Advancing Scientific Digital Twins through AI, Federated Computing and Data," *Future Generation Computer Systems 179, 108312 (2026)*.
- J. Komijani and M. K. Marinkovic, "Normalizing flows for SU(N) gauge theories employing singular value decomposition," *PoS Lattice 2024, 050*.
- J. Komijani and M. K. Marinkovic, "Generative models for scalar field theories: how to deal with poor scaling?" *PoS Lattice 2022, 019*.

Lattice QCD

- R. Aliberti *et al.*, "The anomalous magnetic moment of the muon in the Standard Model: an update," *Phys. Rept. 1143, 1–158 (2025)*.
- A. Bazavov *et al.*, "Up-, down-, strange-, charm-, and bottom-quark masses from four-flavor lattice QCD," *Phys. Rev. D 98, 054517 (2018)*.
- A. Bazavov *et al.*, "B- and D-meson leptonic decay constants from four-flavor lattice QCD," *Phys. Rev. D 98, 074512 (2018)*.
- N. Brambilla, J. Komijani, A.S. Kronfeld and A. Vairo, "Relations between Heavy-light Meson and Quark Masses," *Phys. Rev. D 97, 034503 (2018)*.
- J. Komijani, "A discussion on leading renormalon in the pole mass," *JHEP 08, 062 (2017)*.
- C. Bernard and J. Komijani, "Chiral Perturbation Theory for All-Staggered Heavy-Light Mesons," *Phys. Rev. D 88, 094017 (2013)*.

Mathematical Physics

- J. Komijani, "First-order nonlinear eigenvalue problems involving functions of a general oscillatory behavior," *J. Phys. A: Math. and Theor. 54, 465202 (2021)*.
- C.M. Bender, A. Fring and J. Komijani, "Nonlinear Eigenvalue Problems," *J. Phys. A: Math. and Theor. 47, 235204 (2014)*.

Full list of publications and conference proceedings: available here.